Name: ______________________________ **Unit 6 Test - Modules 13 & 14**

1. ________ **MULTIPLE CHOICE**: Which **exponential function** passes through the points (0, 21) and (1, 3)?

Ⓐ $y = 7\left(\frac{1}{3}\right)^x$

Ⓒ $y = 7\left(\frac{1}{21}\right)^x$

Ⓑ $y = 21\left(\frac{1}{7}\right)^x$

Ⓓ $y = 21\left(\frac{3}{7}\right)^x$

2. ________ **MULTIPLE CHOICE**: What is the **third term** in the geometric sequence $f(n) = 2(0.8)^{n-1}$?

Ⓐ 4.8

Ⓒ 1.28

Ⓑ 2.56

Ⓓ 1.024

3. ________ **MULTIPLE CHOICE**: What is the **recursive rule** for the geometric sequence $f(n) = 13(1.3)^{n-1}$?

Ⓐ $f(1) = 13,\ f(n-1) = 1.3 \cdot f(n),\ n \geq 2$

Ⓑ $f(1) = 1.3,\ f(n-1) = 13 \cdot f(n),\ n \geq 2$

Ⓒ $f(1) = 13,\ f(n) = 1.3 \cdot f(n-1),\ n \geq 2$

Ⓓ $f(1) = 1.3,\ f(n) = 13 \cdot f(n-1),\ n \geq 2$

4. ________ **MULTIPLE CHOICE**: Andrew began with $20 in savings and increased his savings by $10 every month. Lindsay began with $5 in savings and doubled her savings every month. After Andrew and Lindsay each saved for **7 months, who saved more money**?

Ⓐ Andrew, because positive linear functions grow faster than exponential growth functions.

Ⓑ Andrew, because exponential growth functions eventually grow faster than positive linear functions.

Ⓒ Lindsay, because positive linear functions grow faster than exponential growth functions.

Ⓓ Lindsay, because exponential growth functions eventually grow faster than positive linear functions.

5. ________ **MULTIPLE CHOICE**: The first five terms of a sequence are shown in the table below. What is the **recursive rule** for the sequence?

n	1	2	3	4	5
$f(n)$	0.5	1	2	4	8

Ⓐ $f(1) = 2, f(n) = 2 \cdot f(n-1), n \geq 2$
Ⓑ $f(1) = 2, f(n) = 0.5 \cdot f(n-1), n \geq 2$
Ⓒ $f(1) = 0.5, f(n) = 2 \cdot f(n-1), n \geq 2$
Ⓓ $f(1) = 0.5, f(n) = 0.5 \cdot f(n-1), n \geq 2$

6. ________ **MULTIPLE CHOICE**: What is the **fifth term** in the sequence 8, 20, 50, ...?

Ⓐ 125
Ⓑ 250
Ⓒ 312.5
Ⓓ 781.25

7. Place an **X** in the table to show whether each sequence is **arithmetic** or **geometric**.

	Arithmetic	Geometric
78, 69, 60, 51, 42, ...		
4, 12, 36, 108, 324, ...		
−0.6, 0.2, 1, 1.8, 2.6, ...		
225, −22.5, 2.25, −0.225, 0.0225, ...		

8. What is the **common ratio** of the sequence 81, 27, 9, 3, 1, ...? ________________

9. If $f(x) = 625(0.8)^x$, what is the value of **f(4)**? ________________

10. A geometric sequence has a first term of 23 and a common ratio of 7.5. What is the **explicit rule** for this sequence?

$f(n) =$ ______________________________